Synopsis

Project Title

PREDICTIVE ANALYTICS FOR URBAN WATER DEMAND

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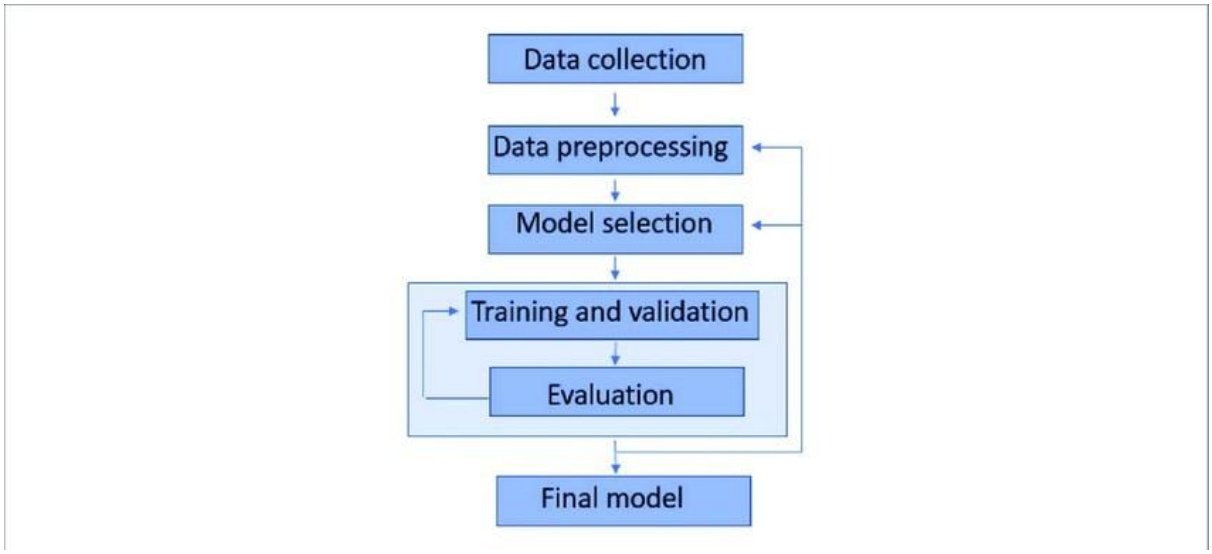
- **Problem Description: -**

In today's rapidly growing urban areas, managing water efficiently has become very important for sustainable living. Daily water usage depends on several factors such as the size of the household, seasonal changes, and the presence of garden areas. However, without proper prediction, it becomes difficult for authorities and households to plan water usage effectively, which may lead to wastage or shortage.

This project focuses on using machine learning to predict daily water consumption in urban households. A dataset will be prepared by collecting information like household size, seasonal index, and garden area. After carefully cleaning the data to remove missing or incorrect values, feature scaling will be applied to ensure consistency in data.

A Linear Regression model will then be used to estimate the daily water usage in liters. The model's performance will be evaluated using residual analysis to understand prediction errors. This approach not only helps in better water management but also promotes awareness about responsible usage. Overall, this project aims to provide a simple yet effective solution for smarter and more sustainable water resource planning.

- **Steps of Implementation :-**



Data Collection	The objective of this project is to analyze and predict urban water demand using machine learning techniques. Various influencing factors such as climate variations, population growth, and historical usage data are studied to understand demand behavior

Data Preprocessing	Data is collected and organized using tools like Python, NumPy, and Pandas. Important attributes such as rainfall, temperature, water consumption, and seasonal trends are included to build a meaningful dataset.

Exploratory Data Analysis (EDA)	The dataset is examined using graphs and statistical methods. Visualization techniques like regression and residual plots are used to identify patterns, relationships, and anomalies in water usage.
Data Cleaning	The dataset is refined by handling missing values, removing inconsistencies, and formatting it properly. Scaling techniques are applied to ensure uniformity among all features.
Feature Selection	Significant features that influence water demand are selected, while irrelevant data is removed. This improves model efficiency and reduces unnecessary complexity.
Model Training	The dataset is divided into training and testing sets. A suitable machine learning model, such as Linear Regression, is trained to learn from historical data.
Performance Analysis	The model is evaluated using test data. Graphical analysis and error measurement techniques are used to determine prediction accuracy and reliability.

Final Deployment	The best-performing model is implemented to predict future water demand. This helps in efficient planning and management of water resources.

• Expected Modules / Libraries:

1. **NumPy:** Used for numerical operations.
2. **Pandas:** Used for data handling and preprocessing.
3. **Scikit-learn:** Used for StandardScaler and Linear Regression.
4. **Matplotlib/Seaborn:** Used for visualization and analysis.

• Expected Evaluation Metrics and Plots:

1. **Accuracy:** Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted instances to total instances. It is useful for understanding the general performance of the classification model.
2. **Precision:** Precision measures how many of the predicted congested cases actually correct. It is important to avoid false positives in congestion prediction.
3. **Recall:** Recall measures how many actual congested cases are correctly identified by the model. It is crucial because missing congestion (false negatives) can lead to poor network performance.
4. **F1-Score:** F1-Score is the harmonic mean of precision and recall. It provides a balanced evaluation when both false positives and false negatives are important.
5. **Confusion Matrix:** A confusion matrix provides a detailed breakdown of correct and incorrect predictions. It helps in understanding model performance in terms of true positives, false positives, true negatives, and false negatives.
6.

• Required Topics from the Subject:

1. **Strings**: Used to handle and process textual data such as column names and dataset labels.
2. **File Handling**: Helps in reading datasets (CSV files) and saving processed outputs.
3. **Libraries (Python Modules):**
   a) **NumPy** for numerical operations
   b) **Pandas** for data manipulation
   c) **Scikit-learn** for StandardScaler and Linear Regression
4. **Data Structures (Arrays/Data Frames):** Used to store and manage dataset efficiently.
5. **Basic Statistics:** Helps in understanding mean, variance, and error calculations.

• Platform Required:

1. **PROGRAMMING LANGUAGE:** Python 3.x
2. **IDE:** Jupyter Notebook / Python IDE / Visual Studio Code
3. **OPERATING SYSTEM:** Windows, Linux or macOS

• **Books and Link Sources:**

1. Python for Data Analysis by Wes McKinney
2. Scikit-learn Official Documentation – https://scikit-learn.org
3. Kaggle (Datasets & Learning) – https://www.kaggle.com
4. W3Schools Python Tutorial – https://www.w3schools.com/python/